Order Flow & Pre-Move Detection System

Project Brief v2 — 18-Feature Edition

PRIMARY OBJECTIVE

Build a system to **pre-detect stocks that will move +/- 5% intraday** (configurable) by identifying common characteristics these stocks exhibit **before the move happens** — using tick-by-tick trade data and 20-level market depth (DOM) across 500 NSE stocks.

> **Key question:** Do stocks that move 5%+ show identifiable order flow / microstructure signals in the minutes before the move — that non-moving stocks do NOT show?

FEATURE SET — 18 FEATURES (computed every 1 minute per stock)

Features are grouped into 4 categories. All are computed from raw ticks + DOM snapshots. The 5% label threshold is a configurable parameter throughout.

Group A — Tick / Trade Flow (6 features)

#	Feature Name	Description	Source
1	delta_1m	Buy volume - Sell volume in last 1 min	Tick
2	delta_5m	Buy volume - Sell volume in last 5 min	Tick
3	volume_burst	Current 1m volume / rolling 20-period avg volume	Tick
4	aggressor_ratio	Aggressive buy trades / total trades (1m window)	Tick
5	trade_count_burst	Number of trades in last 1m vs rolling avg	Tick
6	large_trade_ratio	Trades above 10x median size / total trades	Tick

Group B — Market Depth / DOM (6 features)

#	Feature Name	Description	Source
7	imbalance_top5	(BidQty - AskQty) / (BidQty + AskQty) — top 5 levels	DOM
8	imbalance_top10	Same formula for top 10 levels	DOM
9	imbalance_top20	Same formula for all 20 levels	DOM
10	spread	Best ask price - best bid price (in ticks)	DOM
11	depth_drop_bid	Change in total bid qty from last snapshot to current	DOM
12	depth_drop_ask	Change in total ask qty from last snapshot to current	DOM

Group C — Price Derived (3 features)

#	Feature Name	Description	Source
13	vwap_distance	(Current price - VWAP) / VWAP — how stretched from VWAP	Tick
14	volatility_5m	Std dev of tick prices in last 5 min	Tick
15	price_acceleration	Rate of change of price change (second derivative)	Tick

Group D — Microstructure / Advanced (3 features)

#	Feature Name	Description	Source

16	iceberg_score	Executed qty / displayed qty at touched level — detects hidden orders	DOM+Tick
17	order_cancel_rate	Orders cancelled vs placed in last 1m window (spoofing proxy)	DOM
18	bid_replenishment_rate	How fast bid qty refills after being consumed (absorption proxy)	DOM

Implementation note: Features 1–15 are straightforward to compute. Features 16–18 are more complex but capture the most valuable institutional signals (icebergs, spoofing, absorption). Build 1–15 first, add 16–18 in iteration 2.

FEATURE TABLE OUTPUT SCHEMA (features.parquet)

One row per stock per minute. Example row:

Column	Type	Example
ts	datetime	2026-06-05 10:31:00
symbol	string	TITAGARH
delta_1m	float	-42500
delta_5m	float	-180000
volume_burst	float	3.7
aggressor_ratio	float	0.34
trade_count_burst	float	2.1
large_trade_ratio	float	0.08
imbalance_top5	float	0.72
imbalance_top10	float	0.61
imbalance_top20	float	0.45
spread	float	0.05
depth_drop_bid	float	-85000
depth_drop_ask	float	12000
vwap_distance	float	-0.008
volatility_5m	float	0.23
price_acceleration	float	-0.003
iceberg_score	float	8.2
order_cancel_rate	float	0.41
bid_replenishment_rate	float	0.87
label	string	LONG / SHORT / NO_TRADE

LABEL GENERATION

For each feature row, look forward in time and assign a label:

```
future_return = (max_price_in_next_2h - current_price) / current_price

if future_return > THRESHOLD: label = LONG
elif future_return < -THRESHOLD: label = SHORT
else: label = NO_TRADE

THRESHOLD = 0.05 # configurable — try 0.03, 0.05, 0.07
```

Do NOT manually select stocks. Use all 500. Let future_return decide what is LONG/SHORT/NO_TRADE. The algorithm discovers what separates the movers — you don't need to guess.

BUILD ORDER & TIMELINE

When	File / Task	What to Do
Day 1	feature_factory.py	Load 1 day, 500 stocks. Compute features 1–15. Output features.parquet. Validate values are
Day 2	label_generator.py	Add LONG/SHORT/NO_TRADE labels to every row. Check label distribution (expect mostly N
Day 3–4	backtester.py	Run simple stats: when imbalance_top10 > 0.7, what % of time was label=LONG? Repeat for
Day 5	Features 16–18	Add iceberg_score, order_cancel_rate, bid_replenishment_rate. These are harder — tackle af
Week 2	Scale up	Expand to 5 days, then 10 days. Fix any bugs found at scale.
Week 3+	train_model.py	Only if backtester shows at least 1–2 features have predictive power. Use LightGBM.

ML MODEL (Phase 4 — after validation)

Recommended model: **LightGBM**. Target: predict LONG / SHORT / NO_TRADE.

```
import lightgbm as lgb
model = lgb.LGBMClassifier(n_estimators=500, learning_rate=0.05)
model.fit(X_train, y_train)
print(model.feature_importances_) # which of the 18 features matter most
```

The feature importance output will tell you which of the 18 features actually predict 5%+ moves. This guides further feature engineering and strategy design.

KEY PRE-MOVE SIGNALS TO LOOK FOR

Absorption: Large sell/buy orders hit the book but price barely moves — hidden counterparty absorbing. Mapped to: bid_replenishment_rate, delta divergence.

Iceberg Orders: DOM shows small qty, but far more executes and level refills. Mapped to: iceberg_score.

Spoofing: Large order appears → price reacts → order disappears. Mapped to: order_cancel_rate, depth_drop.

Queue Imbalance Extreme: imbalance_top10 > 0.7 or < -0.7 → price frequently follows direction.

Volume Burst + Delta Alignment: volume_burst > 3x AND delta_5m strongly positive/negative → directional conviction.

Liquidity Vacuum: depth_drop_bid or depth_drop_ask shows sudden thinning → price accelerates.

Cross-Stock Lead-Lag: One sector stock moves → correlated stocks follow. Build after single-stock features are working.

IMPORTANT RULES

- 5% threshold is a parameter — make it configurable from day 1.
- Use ALL 500 stocks. Never hand-pick winners/losers — introduces bias.
- Build and validate on 1 day before scaling to weeks of data.
- Features 1–15 first. Features 16–18 are advanced — add in iteration 2.
- If no feature shows predictive power in the backtester, don't proceed to ML.

- The edge is microstructure — not candlestick patterns or technical indicators.

Brief v2 — 18 features across 4 groups: Tick Flow, DOM, Price Derived, Microstructure. Feed this to your AI agent for step-by-step implementation guidance.